Week 12 Applied Class

Machine Learning – Part 3

Exercise 1: Clustering

A music streaming service wants to understand listening patterns across its users. It has collected information such as the number of listening sessions per week, average session duration, and the proportion of time spent on different music genres. However, the company has not assigned users to any predefined categories. Its analysts propose discovering groups of users with similar listening behaviour before designing new recommendation features.

(a) Explain why unsupervised learning, rather than supervised classification, is appropriate for this task.

(b) In this context, what would a cluster represent? Explain the meaning of high intra-cluster similarity and low inter-cluster similarity.

(c) Two analysts obtain different groupings of the same users. Does this necessarily mean one result is incorrect? Explain briefly.

(d) Give one reason why the choice of features or similarity measure could change the discovered groups.

Exercise 2: K-means Clustering

Consider the following one-dimensional dataset: X={1,2,4,5,7,8}

We apply K-means clustering with (K=2), using Euclidean distance.

(a) Run K-means with initial centroids (c_1=1) and (c_2=2). Repeat the assignment and centroid update steps until convergence.

(b) Repeat K-means with different initial centroids (c_1=1) and (c_2=8).

(c) Compare the final clusters from (a) and (b). Calculate the total within-cluster squared error for each result. What does this tell us about the effect of centroid initialisation?

Exercise 3: Naïve Bayes – Learning Probabilities

A campus support team labels incoming messages as Urgent (U) or Routine (R). It records whether each message contains the words "ASAP" and "deadline". The following six labelled messages form a small training dataset. The team wants to estimate probabilities for a Naïve Bayes classifier.

Message	Class	ASAP	Deadline
1	U	Yes	Yes
2	U	Yes	No
3	U	No	Yes
4	R	Yes	No
5	R	No	No
6	R	No	Yes

(a) Use Maximum Likelihood Estimation (MLE) to calculate the class priors Pr(U) and Pr(R).

(b) Use MLE to estimate the four conditional probabilities Pr(ASAP = Yes | U), Pr(Deadline = Yes | U), Pr(ASAP = Yes | R), and Pr(Deadline = Yes | R). Show the counts used for each estimate and use the given variable names.

(c) Using these MLE estimates and the Naïve Bayes conditional independence assumption, predict the class of a new message with ASAP = Yes and Deadline = Yes. Write the general Naïve Bayes formula using the given variable names before substituting numbers.

(d) If a feature value never occurs in one class in the training data, what problem can this cause? Briefly explain how ELE smoothing addresses it.

Exercise 4: Naïve Bayes

A maintenance team uses Naïve Bayes to classify a machine as Needs Inspection (I) or Normal (N). The classifier uses two binary observations: High noise and High vibration. The following probabilities have already been estimated from historical data. Assume the two observations are conditionally independent given the class.

Probability	Inspection (I)	Normal (N)
Prior Pr(C)	0.4	0.6
Pr(Noise = High | C)	0.75	0.25
Pr(Vibration = High | C)	0.80	0.20

A newly observed machine has High noise and High vibration.

(a) Explain what conditional independence means for these two observations.

(b) Write the Naïve Bayes formula using Noise, Vibration and class C. Then calculate the unnormalised posterior values for I and N for a machine with Noise = High and Vibration = High. Predict its class.

(c) Write the formula for calculating the posterior probability before substituting numbers. Calculate Pr(I | Noise = High, Vibration = High).

(d) Is your classification reasonable, even though Normal has the larger prior probability? Explain using the likelihoods and the calculated unnormalised posterior values. Identify one assumption or limitation that could make the prediction less reliable.